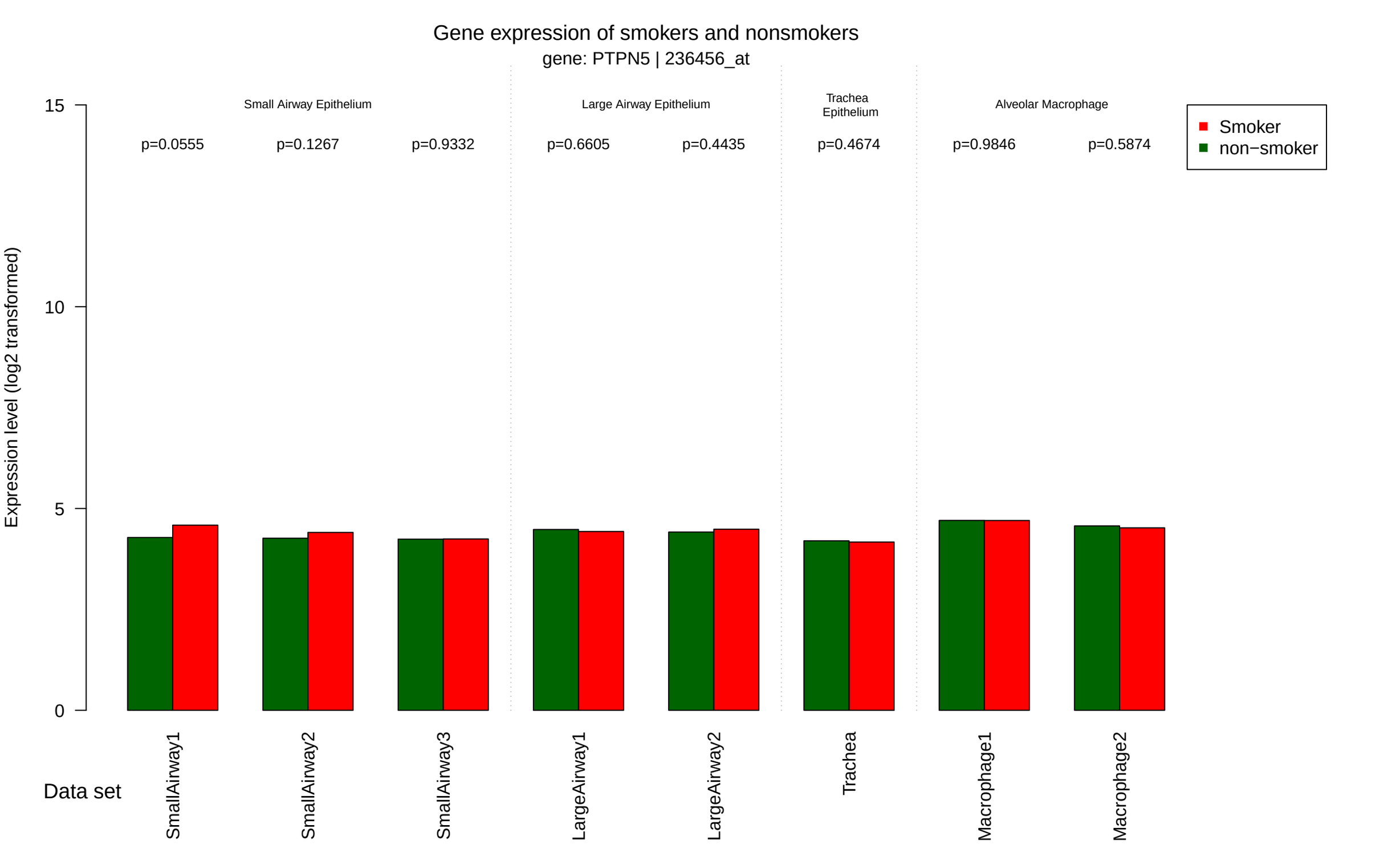
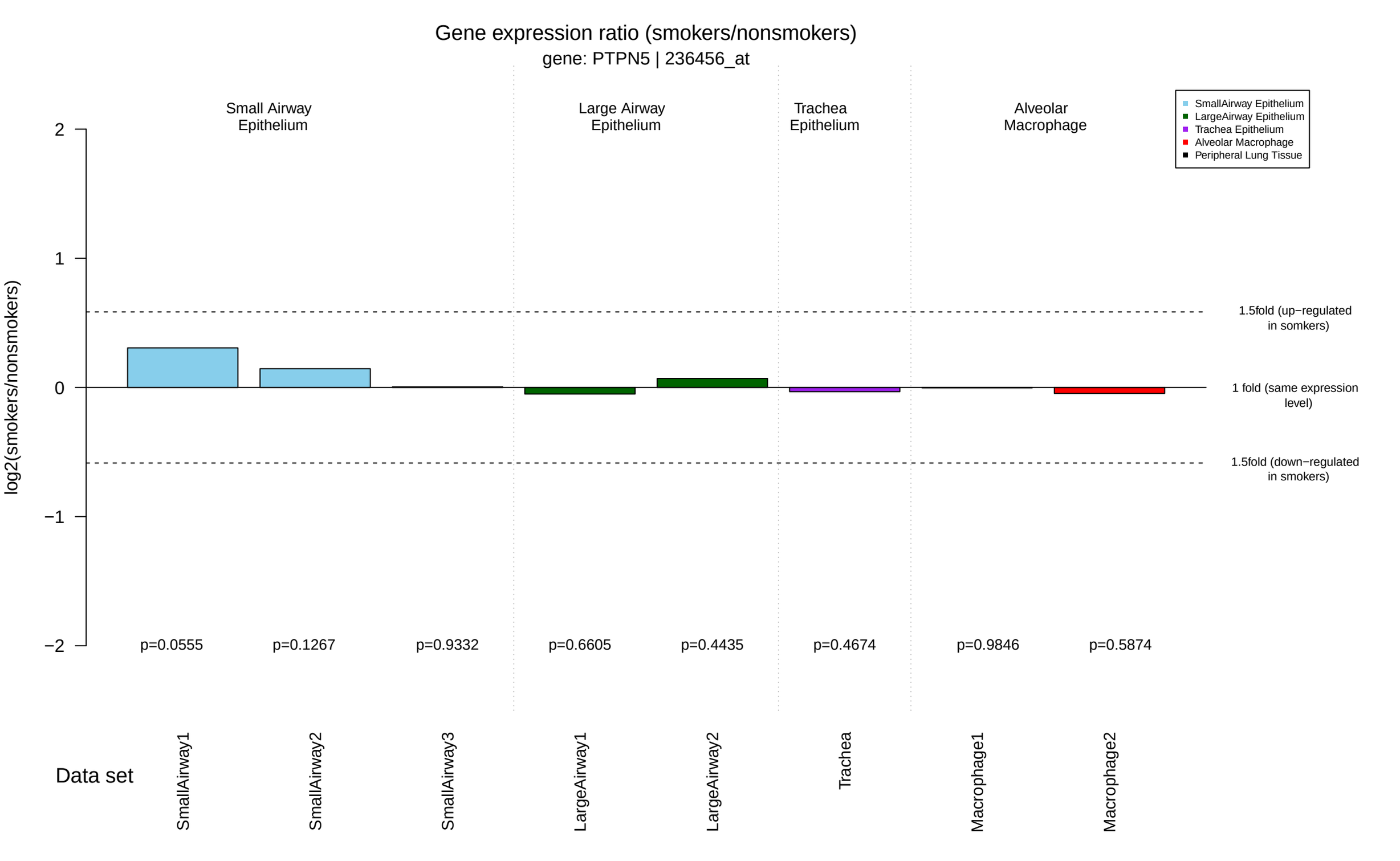
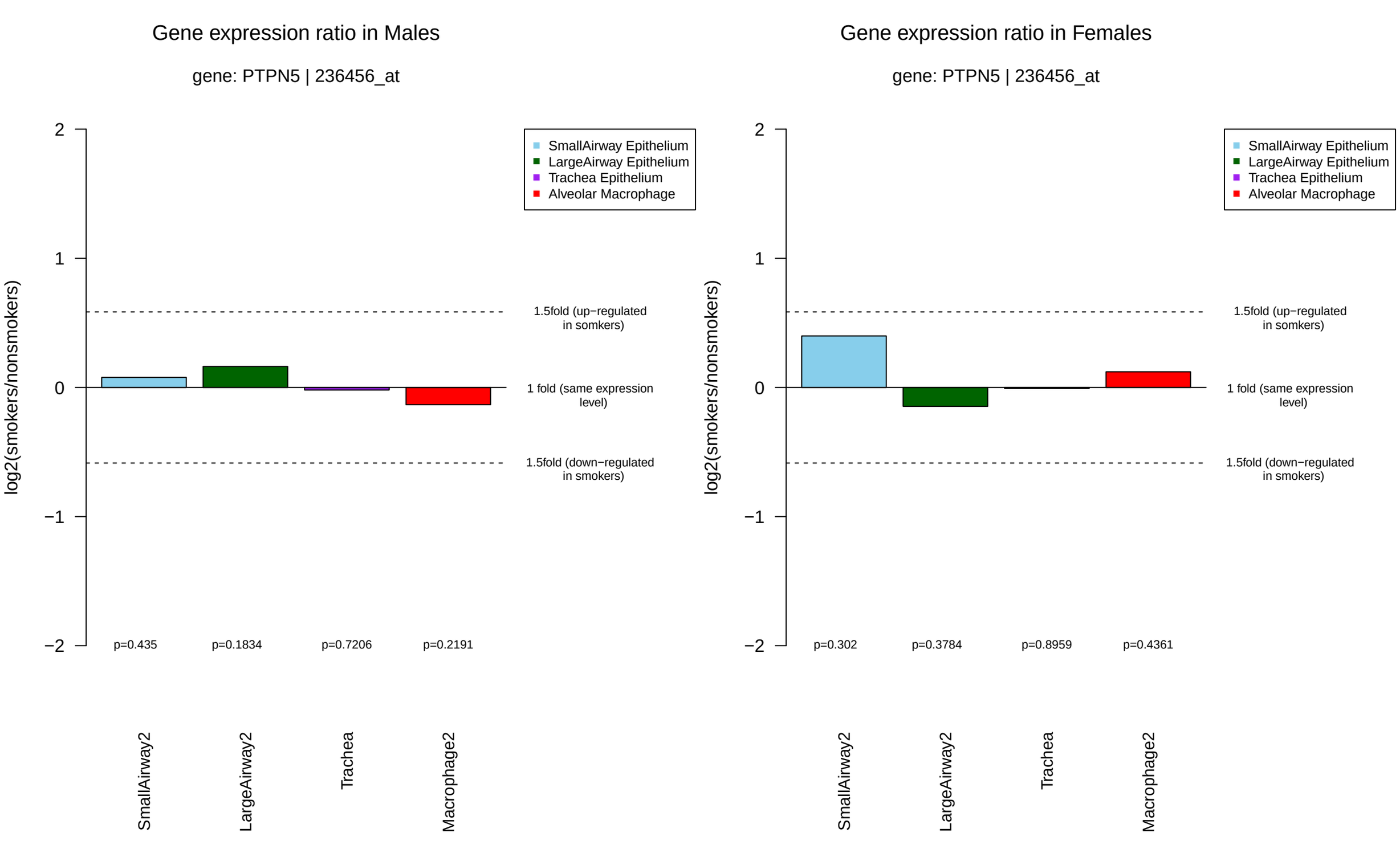
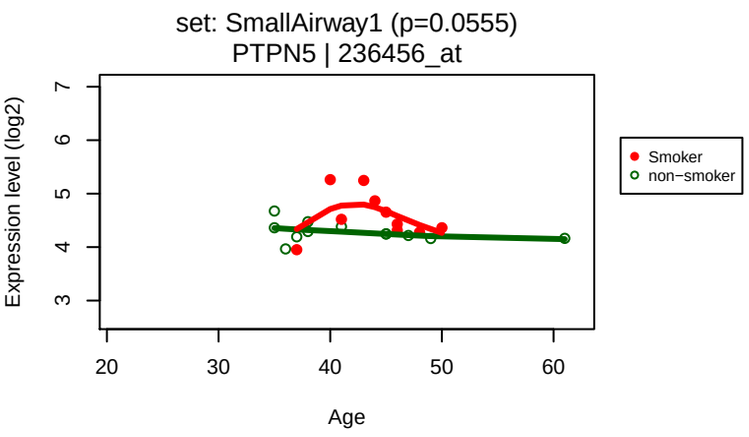
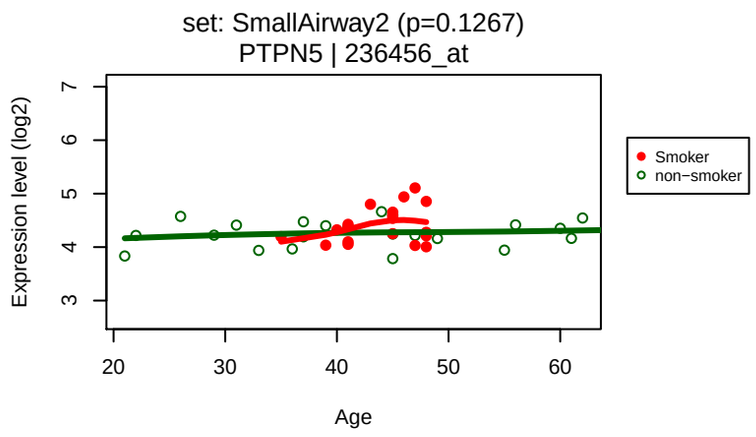
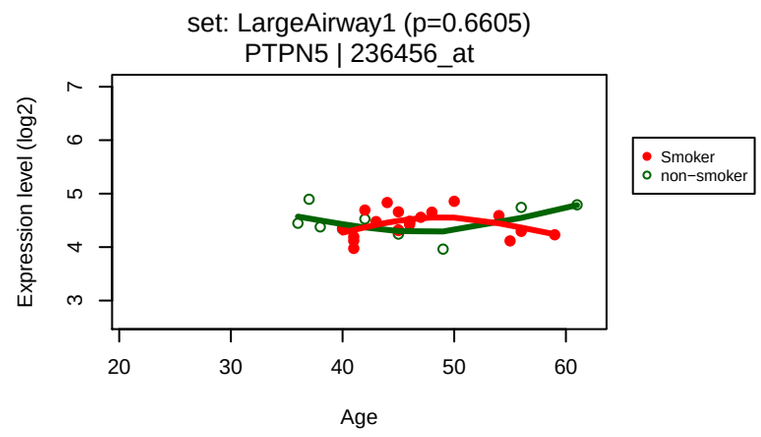

Gene expression of smokers and nonsmokers
gene: PTPN5 | 236456_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0555
p=0.1267
p=0.9332
p=0.6605
p=0.4435
p=0.4674
p=0.9846
p=0.5874
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

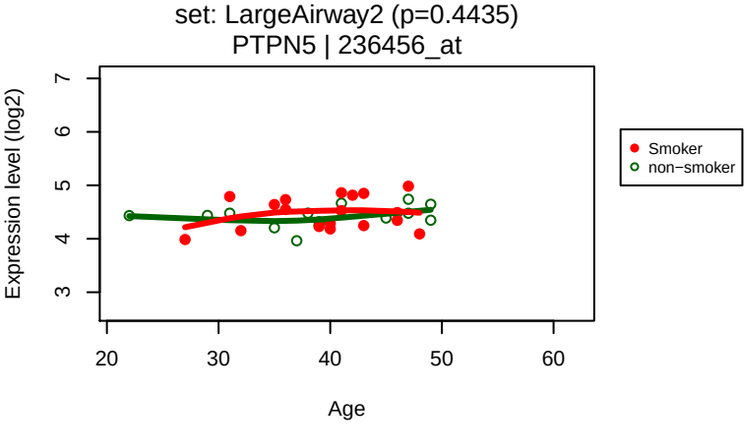

Gene expression ratio (smokers/nonsmokers)
gene: PTPN5 | 236456_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0555
p=0.1267
p=0.9332
p=0.6605
p=0.4435
p=0.4674
p=0.9846
p=0.5874
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

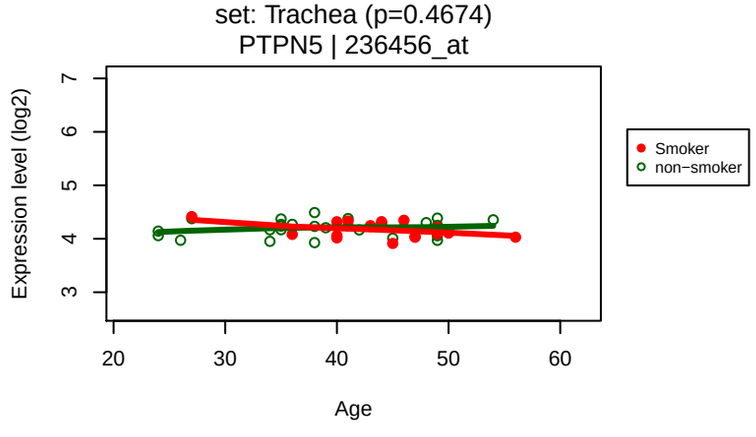

Gene expression ratio in Males
gene: PTPN5 | 236456_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.435
p=0.1834
p=0.7206
p=0.2191
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN5 | 236456_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.302
p=0.3784
p=0.8959
p=0.4361
SmallAirway2
LargeAirway2
Trachea
Macrophage2

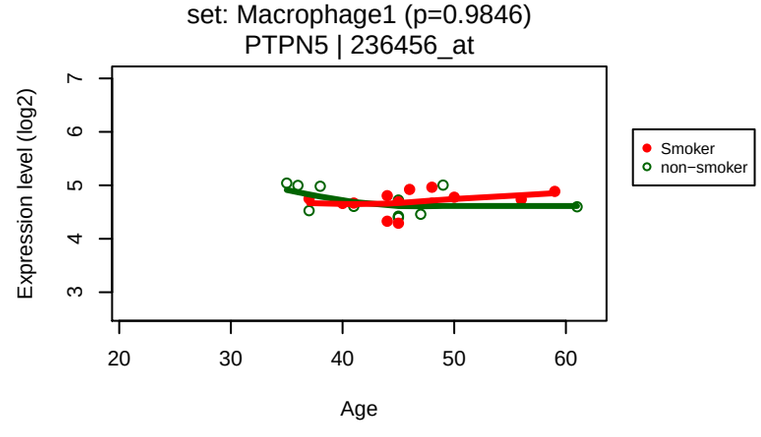

set: SmallAirway1 (p=0.0555)
PTPN5 | 236456_at
Expression level (log2)
Age
Smoker
non−smoker

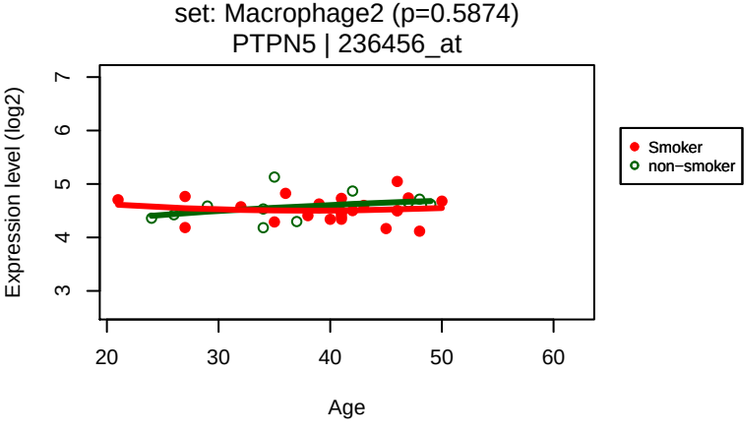

set: SmallAirway2 (p=0.1267)
PTPN5 | 236456_at
Expression level (log2)
Age
Smoker
non−smoker

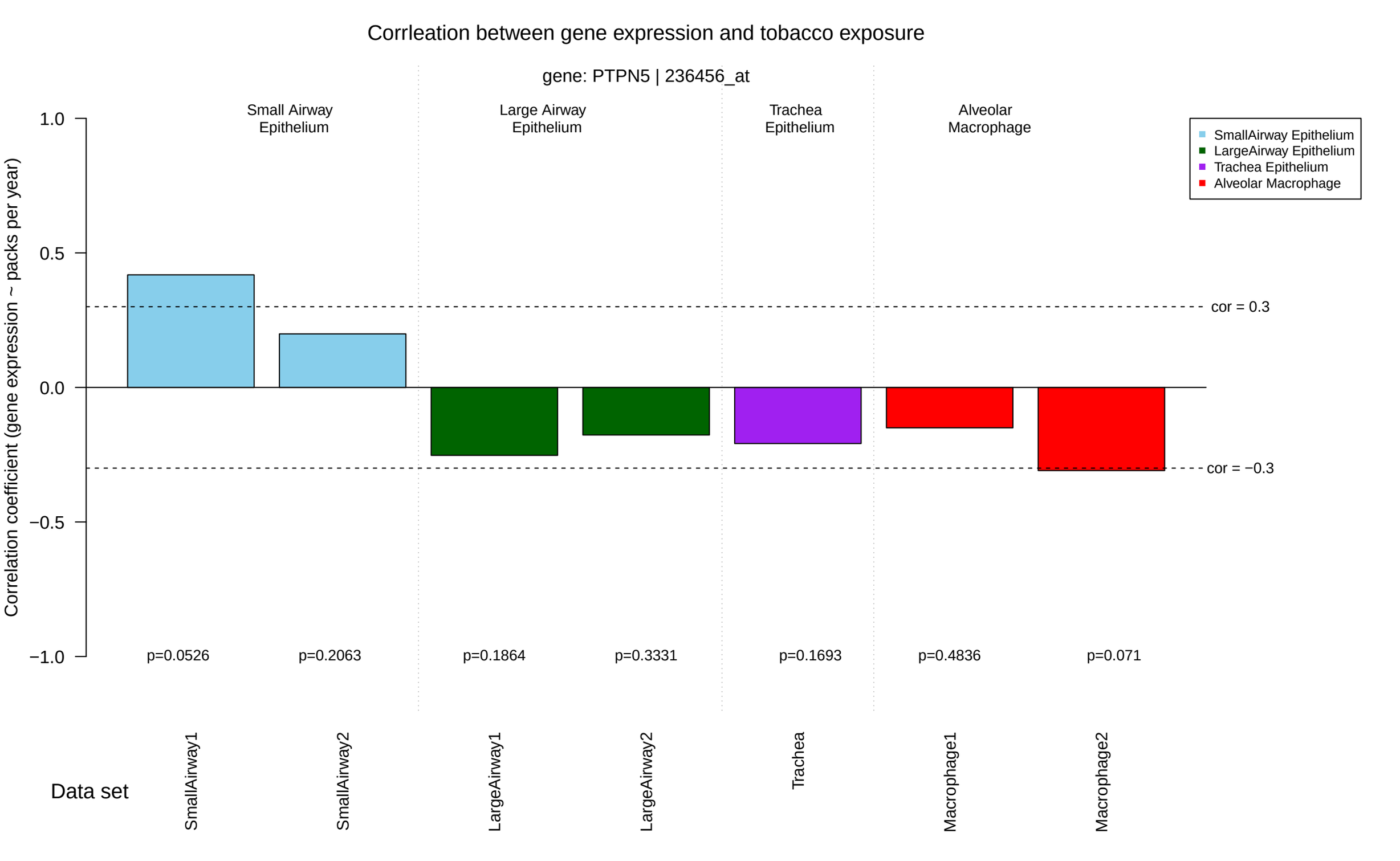

set: LargeAirway1 (p=0.6605)
PTPN5 | 236456_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4435)
PTPN5 | 236456_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4674)
PTPN5 | 236456_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9846)
PTPN5 | 236456_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5874)
PTPN5 | 236456_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN5 | 236456_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0526
p=0.2063
p=0.1864
p=0.3331
p=0.1693
p=0.4836
p=0.071
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2